SSLC

PHYSICS

Model Questions
for D+ grade

D+ Kollam

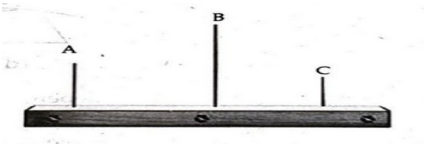
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1. Sound Waves

1.Number of oscillations made in one second of a wave motion is known as.....
(Frequency, amplitude, period, wavelength) (1)

2.



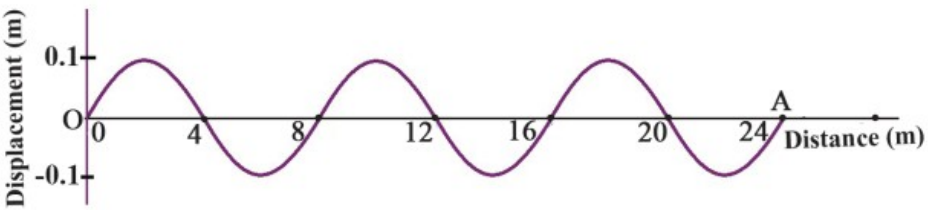
A, B, C are three metallic wires. When they are made to vibrate, which one vibrates with more frequency ? (1)

3. When a tuning fork of frequency 256Hz excited and its stem is kept on a table, sound is heard.
a) what is the frequency with which the table vibrates? (1)
b) what is the name of this phenomenon ? (1)

4. When tuning forks of different frequencies are excited and their stems are placed on a table , sound is heard loudly at a particular situation. What phenomenon is behind this? (1)

5. Some statements related to Wave Motion are given below. Classify them into those of transverse wave and longitudinal wave. (2)
a) Particles in the medium vibrate perpendicular to the direction of propagation of wave
b) particles in the medium vibrate parallel to the direction of propagation of wave
c) compressions and rarefactions are formed
d) Crests and troughs are formed

6. Observe the figure showing a wave motion.



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a) What is the amplitude of this wave
b) Write the wavelength (2)

7. Which waves cause tsunami and earthquake? (1)

8. Which of the following frequency cannot be heard by humans (1)
a)50Hz b)2000Hz c)30000 Hz d)1000Hz

Answer key

1.Frequency
2.C
3.a 256 Hz
b Forced vibration
4 Resonance
5 a Transverse wave
b Longitudinal wave
c Longitudinal wave
d Transverse wave
6. a) 0.1 m
b) 8 m
7. Seismic waves
8. c 30000 Hz

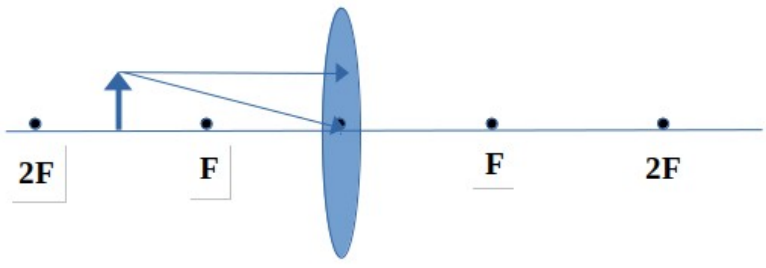
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2 . Lenses

1. Which of the following is the midpoint of lens (1)
(Centre of curvature, optic centre, principal focus, optic axis)

2. Which lens has negative focal length. (1)

3. Observe the diagram



a) Complete the ray diagram of image formation (2)
b) Write any two characteristics of image (1)

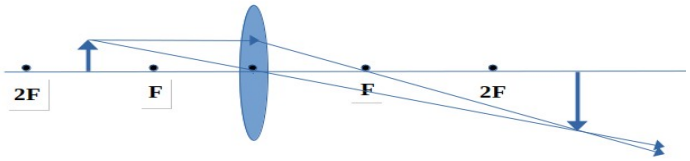
4. Write the unit of power of lens (1)

5. Which lens always forms diminished, virtual image? (1)

6. Find out the correct statements from the following (2)
a) the focal length of objective of compound microscope is more
b) the focal length of objective of telescope is more
c) the focal length of objective of compound microscope is less
d) the focal length of objective of telescope is less

Answer key

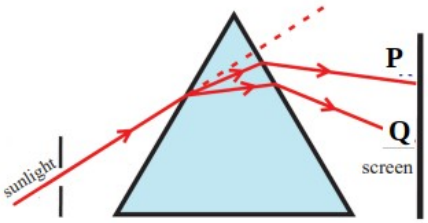
1 Optic centre
2 Concave lens
3 a)



b) Magnified, Real , Inverted (any 2)
4 Dioptre (D)
5 Concave lens
6 b) the focal length of objective of telescope is more
c) the focal length of objective of compound microscope is less

3 .The world of colours and vision

1. White light passing through a prism is depicted
a) What do the colours P and Q represent (1)
b) What is the name of the phenomenon related to this?(1)



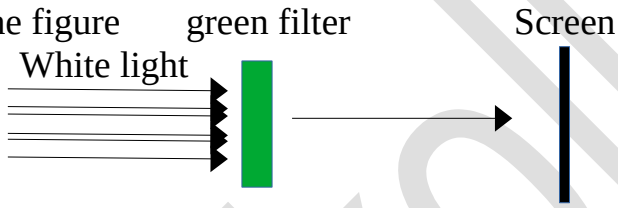
2. Choose the phenomenon behind the formation of a rainbow (1)
(Tyndall effect, dispersion, scattering, persistence of vision)

3. Fill in the blanks : (2)
a) Red+ green = ...
b) Red+Blue = ...
c) Blue +Green = ...
d) Red+green+Blue = ...

4. Which is the complementary colour of yellow? (1)
(Green,red, blue,violet)

5. When a fan rotates fast, it's leaves are not seen distinctly. Name the phenomenon behind this. (1)

6. Observe the figure

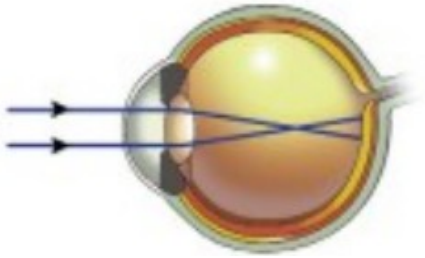


What colour is seen on the screen (1)

7a). In which colour is the red flower seen in red light ?
b) In which colour is the red flower seen in blue light? (2)

8. Choose the phenomenon suitable for each of the following situations from the options given in the bracket (2)
(Power of accommodation, persistence of vision, scattering, Tyndall effect)
a) the path of sunlight is visible in misty morning
b) Newton's colour disc appears white when rotated fast
c) rising and setting sun appears red
d) a person can see the far objects and near objects clearly

9. Observe the picture and answer the following questions



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a) what is the name of this defect of eye?
b) what are the reasons for the defect .
c) which lens can be used to rectify this defect? (3)

Answer key

1 a) P - Red, Q - Violet
b) Dispersion of light
2 Dispersion
3 a) Yellow
b) Magenta
c) Cyan
d) White
4 Blue
5 Persistence of vision
6 Green
7 a) Red
b) Dark
8 a) Tyndall effect
b) Persistence of vision
c) Scattering
d) Power of accommodation
9 a) Myopia (Short sightedness)
b) Size of the eyeball is larger, Power of the lens is more
c) Concave lens

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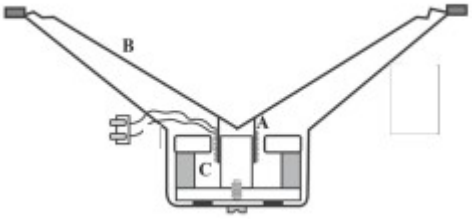
4. Magnetic effect of electric current

1. Name the rule used to find the direction of magnetic field around a current carrying conductor (1)

2. If current flows anticlockwise at one end of a solenoid, which pole is formed there? (1)

3. Suggest two methods to increase the strength of magnetic field of a current carrying solenoid (2)

4. Observe the figure



a) write the name of the device
b) identify the parts A, B, C
c) write the working principle
d) what is the energy change in this device
e) write the name of another device that works on this principle (4)

Answer key

1 Right hand thumb rule
2 North pole
3
- Increase the number of turns of solenoid
- Increase the intensity of current
- Use soft iron as the core
- Increase the area of cross section of the core (any 2)

4
a) Moving coil loudspeaker
b) A : voice coil
B : paper diaphragm
C : field magnet
c) Motor principle
d) Electrical energy to Sound energy
e) Fan/Electric motor/Mixie

5. Electric energy consumption and conservation

1.State joule's law. Represent it mathematically . (3)

2. Which of the following is not the equation of power (1)
(**$P = V \times I$, $P = I^2R$, $P = V^2 / R^2$, $P = V^2 / R$**)

3. a) What is the name of the device used to measure electric energy consumed in houses
b) what is the commercial unit of electrical energy (2)

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4. An electrical appliance is marked 800 W, 230 V
   a) what do these markings indicate
   b) find out the resistance of this device (3)

5. Write two methods to reduce carbon footprint (2)

6. a) What is Energy crisis?
   b) write two methods to solve it. (3)

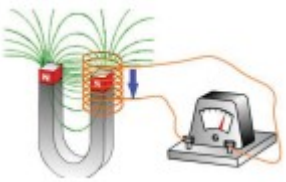
Answer key

1. The quantity of heat produced in a current carrying conductor is directly proportional to the square of the current, the resistance of the conductor and the time of flow of current
$H = I^2Rt$
2. $P = V^2/R^2$
3 a) Watt hour meter
   b) Kilowatt hour (kWh)
4 a) power of the appliance is 800 W at 230 V/ power 800 W and voltage 230 V
   b) $R = V^2/P$
   $= 230 \times 230 / 800$
   $= 66.125\ \Omega$
5.
- Reduce domestic energy consumption
- Use public transport (any 2)

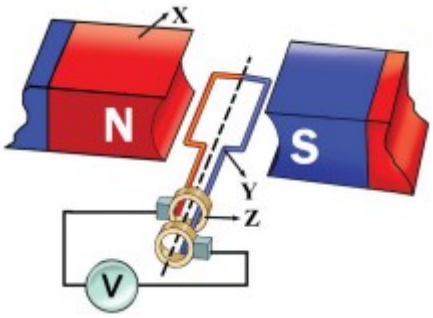
6 a) Increase in demand and decrease in availability of energy
   b) Use solar panel , solar cooker
   Use LED tubes and bulbs

6. Electromagnetic induction in daily life

1. A solenoid connected to a galvanometer is brought to one pole of a magnet, galvanometer needle deflects
a) Name the phenomenon related to this
b) Define it (3)

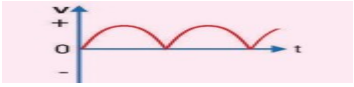


2. Observe the following figure



(a) Identify the device and write down.
(b) write down the main parts of this device.
(c) what is the working principle of this device(3)

3. Match the following suitably. (3)

Graph of current	Source of current
	DC generator
	DC battery
	AC generator

4. At what voltage is electricity generated in power stations in India (1)
(11KV, 11V, 110V, 230V)

5. What type of electricity is generated in the armature of DC generator (1)
(AC, DC, current at constant voltage, none of these)

6. What is the voltage of electricity supplied for household use in our state? (1)
(230V, 230KV, 11KV, 11V)

7. Some statements related to transformers are given below. Classify them into those for step up and step down transformers (2)
a) number of turns in primary is more
b) number of turns in secondary is more
c) Thick wires are used in primary
d)Thin wires are used in primary

8. In a transformer of no power loss, power given to primary is 500 W, what will be power in the secondary? (1)

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9. Safety fuse, MCB, ELCB are commonly used safety measures in house hold circuit.
(a) Which alloy is used to make fuse wire?
(b) Which effect of electricity is used in fuse wire?
(c) What is the peculiarity of fuse wire?
(d) Which device helps to ensure safety of circuit from current leak? (4)

10 (a) Which line in three pin plug is connected to the metallic part of device?
(b) What is the colour of this wire ? (2)

11. (a) Write two precautions to avoid electric shock?
(b) Write two first aid given to a person who get electric shock. (2)

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Answer key

1 a) Electromagnetic induction
b) An emf is induced across a conductor due to change in magnetic flux linked with the conductor

2
a) AC generator
b) Armature, field magnet,slip rings and brush
c) Electromagnetic induction

3.

Graph of current	Source of current
	AC generator
	DC generator
	DC battery

4. 11 KV
5. AC
6. 230 V

7 a Step down transformer
b Step up transformer
c Step up transformer
d Step down transformer
8. 500 W
9. (a) alloy of tin and lead
(b) heating effect
(c) fuse wire has low melting point.
(d) ELCB
10. (a) earth line (E)
(b) green
11 (a)

- Do not operate electric appliances with wet hand
- Do not fly kite near electric line
- do not plug in or plug out when switch is on.
- Do not touch the inside of cable TV adapter (any two)

(b)

- Rub the body to increase body temperature
- Give artificial respiration
- Rub muscles to bring them to normal state.
- start first aid to heart.
-

7. Mechanical advantage in action

1 Write down two advantages of using simple machines. (2)

2. Mechanical advantage of any simple machine is found by : (1)
(Load x effort Effort/Load Load/Effort)

3. Write the principle of a lever. (1)

4. Complete the table (2)

Order of lever	Relative position of fulcrum, load and effort	Mechanical advantage
First order	Fulcrum is in between the load and the effort	Greater than one / equal to one / less than one.
Second order		
Third order		

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5. Draw a schematic diagram of second order lever. Mark the fulcrum, load and effort. (2)

6. Lemon squeezer is a..... order lever. (1)

7. (a) To increase the speed of a vehicle, toothed wheel attached to engine is connected to which toothed wheel?
(small / large)
(b) Which toothed wheel is to be connected while going uphill?
(small / large) (2)

8. The simple machine which has a double inclined plane is..... (1)

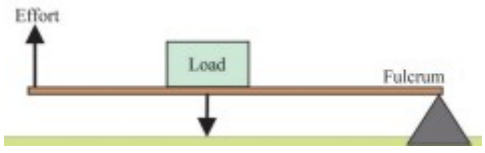
Answer key

1. Changes the magnitude of effect of force.
Changes the direction of applied force.
2. Load / effort
3. Load x Load arm = Effort x effort arm

4.

Order of lever	Relative position of fulcrum, load and effort	Mechanical advantage
First order	Fulcrum is in between the load and the effort	Greater than one / equal to one / less than one.
Second order	Load is in between fulcrum and effort	Greater than one
Third order	Effort is in between load and fulcrum	Less than one

5.



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6. second order
7. (a) small
   (b) large
8. wedge

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